a. **Access Control**

```
START
 INPUT username
 INPUT password
 IF login credentials are correct and active THEN
  DISPLAY "Open access permission"
  ELSE
  DISPLAY "Incorrect username or password"
 ENDIF
END
```

b. **Worker Health and Safety Monitoring**

```
START
READ workers.csv
 INPUT "Do you want to add a worker?"
 IF answer = YES THEN
 INPUT worker information
 CREATE worker ID
 SAVE worker information to workers.csv
 ELSE
DISPLAY worker records and warnings
ENDIF
END
```

c. **Safety Incident Database**

```
START
READ incidents.csv
INPUT "Do you want to add an incident?"
```

IF answer = YES THEN

INPUT incident information

CREATE incident ID

SAVE incident information to incidents.csv

ELSE

DISPLAY incident records and filters

DISPLAY incident analysis

ENDIF

END

d. Equipment Database

START

READ equipment.csv

INPUT "Do you want to add new equipment?"

IF answer = YES THEN

INPUT equipment details

CREATE equipment ID

SAVE equipment details to equipment.csv

ELSE

DISPLAY equipment records and filters

DISPLAY equipment analysis

ENDIF

END

e. Equipment Condition Monitoring

START

READ equipment.csv

INPUT temperature condition

```
IF condition = CRITICAL THEN
DISPLAY "CRITICAL"
ELSE IF condition = WARNING THEN
DISPLAY "WARNING"
ELSE
DISPLAY "NORMAL"
ENDIF
DISPLAY equipment condition message
END
```

f. Equipment Maintenance Monitoring

```
START
READ equipment.csv
CALCULATE equipment availability
DISPLAY equipment that requires maintenance
DISPLAY maintenance records
END
```

g. Risk Assessment

```
START
INPUT likelihood
INPUT consequence
IF likelihood >= 1 AND likelihood <= 5 AND
consequence >= 1 AND consequence <= 5 THEN
risk = likelihood * consequence
DISPLAY risk results
```

```
    DISPLAY action alert
  ELSE
    DISPLAY "Value must be less than or equal to 5"
  END IF
END
```

h. Alerts and Notifications

```
START
  CHECK conditions for all records
  IF threshold condition is met THEN
    DISPLAY ID
    DISPLAY parameter
    DISPLAY value
    DISPLAY status
    DISPLAY recommended action
  ELSE
    DISPLAY "No Alerts"
  END IF
  DISPLAY permitted alerts
END
```

i. Mining Monitoring Dashboard

```
START
  OPEN dashboard
  READ all CSV files
  READ KPIs
```

```
    DISPLAY data
    DISPLAY summary for analysis
END
```

j. Interactive Data Analysis

```
START
INPUT selected question
 DISPLAY answer from datasets
 INPUT download report
 IF download report == "Yes" THEN
    DOWNLOAD report
 END IF
END
```